I. THE PLANT

1. The Plant

a) The Plant - The plant is a living organism which usually consists of root, stem, and leaves. (Latin: planta - a shoot.)
b) The Root - The root is the descending axis of the plant. It takes in water which has mineral salts dissolved in it and frequently fixes the plant to the ground. It bears no leaves or flowers and grows downwards. (Middle English - roote.)
c) The Stem - The stem is the ascending axis of the plant. It bears buds, leaves, flowers and fruit and connects these with the root. It grows upwards, the apex of the stem is called the growing point. (Anglo-Saxon: Stemn - stem of a tree.)
d) The Leaves - The leaves are the laminar expansions, veined and usually green in colour, which proceed from the stem.
e) The Axis of the Plant - The axis of the plant is the central part or longitudinal support around which the organs are arranged.

II. ROOTS

2. Types of Roots

a) Tap Roots or Branched Roots - A tap root is one which descends deeply into the ground and generally produces branches. (Oak, Wallflower, Bean.) (English: tap - to pierce.)

b) Fasciculated Roots - Fasciculated roots are roots which are generally fibrous (threadlike) and all alike. (Grass.) (Latin: fiber - thread)

3. Swollen Roots - In some cases, roots are used for food storage. Then they present swellings in certain regions or are entirely swollen. To distinguish them from one another they are given different names chosen from some familiar object the shape of which they resemble. But as the names for these objects differ in different languages the words adopted are taken from Latin or Greek and each country then uses the same names. So ‘conical’ means similar to a cone; ‘fusiform’ similar to a spindle (the Latin word for spindle is fusus); etc. (English: swell - to increase in volume.)

a) Conical Roots - Conical roots are roots which are broad at the stem and gradually taper towards the tip. (Carrot.) (Latin: conus - a cone.)
b) Napiform Roots - Napiform roots are roots which assume the shape of a turnip. Large and round in the upper part and tapering abruptly below. (Latin: napus - a turnip.)
c) Fusiform Roots - Fusiform roots are roots which taper at each end in the shape of a spindle. (Radish.) (Latin: fusus - a spindle.)
d) Tuberous Roots - Tuberous roots are roots which assume the shape of a tuber. (Dahlia.) (Latin: tuber, from tumere - to swell.)

III. STEM OR TRUNK

4. Types of Stems or Trunks

a) Aerial Stems - Aerial stems are stems which grow above the ground. (Latin: aer - air.)
b) Subterranean or Underground Stems - Subterranean or underground stems are stems which grow beneath the ground. (Latin: sub - under; terra - ground.)

5. Parts and Attachment of Stem

a) Node - A node is the point of attachment of the leaf or leaves to the stem. (Latin: nodus - a knot.)
b) Internode - An internode is the portion of stem between two successive nodes.
c) Axil - An axil is the angle between a branch or leaf and the stem from which it springs.
d) Bud - A bud is a small axillary or terminal protuberance composed of successive layers of rudimental floral or foliage leaves. Buds are therefore undeveloped stems or inflorescences.

6. Subterranean Stems

a) Rhizome - A rhizome is an underground, rootlike stem, which sends up leafy shoots from the upper surface and produces roots from the lower side. (Solomon's Seal, Iris.) (Greek: rizoma - a mass of roots.)
b) Tubers - Tubers are short, fleshy, underground stems bearing very small scale leaves at the axil of which are situated buds: the 'eyes' in the potato. (Potatoes.)
c) Bulb - A bulb is a short thickened stem with a large number of crowded, overlapping fleshy leaves. (Tulip, Onion.) (Latin: bulbus - a bulb.)
d) A corm is a short, fleshy, bulb - lie stem surrounded by a few thin membranes or scale leaves and bearing buds at the summit. (Crocus, Gladiolus.) (Greek: kormos - trunk of a tree shorn of branches.)

7. Aerial Stems - Habit of Growth

a) Erect - An erect stem is a stem which grows upright without any support. (Latin: erectus - having a vertical position.)
b) Procumbent - A procumbent stem is a stem which trails along the earth. (Latin: pro - forward; cumbere - to bend.)
c) Clumbing - A climbing stem is a stem which needs support to go aloft. (English: cleave - to adhere.)

8. Erect Stems

a) Herbaceous - Erect stems are called herbaceous when they are soft and green. They are usually annual. (annual - whose life lasts only one year.) (Latin: herbaceous - grassy.)
b) Shrubby - Erect stems are called shrubby when from a short, hard and woody base several stems arise at or near the ground. (Box, Heather, Gooseberry.) (
c) Woody - Erect stems are called woody when there is a single hard stem, more than ten feet high when grown. It is usually devoid of branches in the lower portion, but bears a group of branches and leaves, or only a bunch of leaves at the top. (Trees.) (Middle English: wode - wood.)

IV. LEAVES

9. Parts of the Leaf

a. The Leaf
b. The Veins - The veins are the portions of the leaf that function like the veins in the human body. They are made of harder material and act as a support to the softer parts of the leaf, forming a kind of skeleton. The larger veins are therefore also called ribs. (Latin: vene - vein, also tube; costa - rib.)
c. The Blade - The blade is the lamina or fully expanded part of the leaf. (Middle English: blad - leaf.)
d. The Margin - The argin is the outer rim of the leaf. (Latin: margo - marginis - a border.)
e. The Petiole - The petiole is the narrow portion which joins the blade to the stem. (Latin: petiolus - a little foot.)
f. The Stipules - The stipules are the leaflets sometimes found at the base of the petiole, having a lateral position. They may be free, or be joined together (adnate) or to the stem, or petiole. (Latin: stipula - a stalk.)
g. The Apex - The apex is the point of the leaf. (Latin: apices - summit.)

10. Venation of Leaf

a. Reticulate - The veins are reticulate when on entering the leaf from the petiole they are continued in the form of one or more ribs that give off branches on either side. These, in their turn, branch off and also produce thinner branches called veinlets so that the whole forms a network. (Latin: reticulatus - resembling network.)
b. Parallel - The veins are parallel when after entering the leaf they run more or less parallel to each other and are united by simple transverse veins. (Palms, Leeks, Grasses.) (Greek: parallelos: para - beside; allelon - one another.)

11. The Blade

a. Simple - A leaf is called simple when the blade consists of a single piece. The argin may be more or less deeply broken, but the incisions never reach the central vein. (Latin: simplex - simple.)
b. Compound - A leaf is called compound when the blade is divided down to the midrib, giving the appearance of many small leaves close together. Each division is called a leaflet and has a small petiole, so that each leaflet looks like a small complete leaf, that they together form one leaf is shown by the bud in the axil of the compound leaf. (Latin: cum - together; ponere - to put.)

12. The Form of Leaves - A

a. Ovate - Ovate is the name given to a leaf which is shaped like the longitudinal section of an egg with the wide part at the base. (Lilac.) (Latin: ovum - egg.)
b. Obovate - Obovate is the name given to a leaf which is shaped like an egg with the wide part at the top. (Woodsorrel.) (Latin: ob - inversely.)
c. Cordate - Cordate is the name given to a leaf which is shaped like a heart. (Deadnettle.) (Latin: cor, cordis - heart.)
d. Obcordate - Obcordate is the name given to a leaf which is shaped like a heart but inverted. (White Clover.)
e. Lanceolate - Lanceolate is the name given to a leaf which is narrow and tapers to a point, with the broadest part at the base and the point at te apex, thus assuming the shape of a lance. (Wallflower.) (Latin: lancea - lance.)
f. Oblanceolate - Oblanceolate is the name given to a leaf which has the shape of a lance, but with the broadest part at the apex and the point at the base.
g. Hastate - Hastate is the name given to a triangular leaf when the angles or lobes at the base spread out resembling the shape of halberd. (Arum or Cuckoo Pint.) (Latin: hasta - spear.)
h. Sagittate - Sagittate is the name given to an elongated triangular leaf, with the two angles at the base prolonged downward like an arrowhead. (Convolvulus, Saggitaria.) (Latin: saggita - arrow.)

13. The Form of Leaves - B

a. Spatulate - Spatulate is the name given to a leaf with a short broad part near the top and long narrow tapering part below it. In form it suggests a spatula or flat ladle. (Daisy.) (Latin: spatula - a spoon.)
b. Reniform - Reniform is the name given to a leaf which has the shape of a kidney. (Ground Ivy.) (Latin: ren, renis - kidney.)
c. Elliptical - Elliptical is the name given to a leaf which has the shape of an ellipse. (Apple.) (Greek: elleipsis - ellipse.)
d. Linear - Linear is the name given to a leaf which is narrow, comparatively long, and uniform in width. It is many times longer than it is wide, somewhat like a ribbon. Sometimes it is so narrow as to suggest a line. (Grasses.) (Latin: linea - line.)

e. Aciculate - Aciculate is the name given to a leaf which is like a needle, slender, narrow and pointed. (Fir.) (Latin: acus - needle.)
f. Orbiculate - Orbiculate is the name given to a leaf which is round. (Marsh Pennyworth.) (Latin: orbis - circle.)
g. Ensiform - Ensiform is the name given to a leaf which is sword shaped, having sharp edges and tapering to a slender point. (Iris.) (Latin: ensis - sword.)
h. Triangular - Triangular is the name given to a leaf which has the shape of a triange. (Beech.) (Latin: tres - 3; angulus - angle.)

14. Further Forms of Leaves

a. Lyrate - Lyrate is the name given to a leaf with a large rounded lobe at the apex and with smaller lobes towards the base. (Latin: lyra - lyre.)
b. Runcinate - Runcinate is the name given to a leaf in which there is a large pointed terminal lobe, the apices of the smaller lobes below are directed downwards. (Dandelion.) (Latin: runcina - a plane.)
c. Peltate - Peltate is the name given to a shield - shaped leaf which has the petiole attached to the lower surface, instead of at the base. (Nasturtium.) (Greek: pelte - shield.)

15. The Margins of Leaves

a. Entire - The margins of leaves are termed entire when there is no incision. (Lilac.) (Latin: interger: in - non; tangers - to touch, untouched.)
b. Dented - The margins of leaves are termed dented when the incision is slight. (Birch.) (Middle English: dint - angular incision.)

16. When the Margins Have Deeper Incisions They Take on a Double Name - The Pinnately - The reticulate unicostate leaves with incisions deeper than the ones called dented, prefix to the respective terms the word pinnati or pinnately. This is because the divisions resulting from the incisions are disposed along the midrib like the barbs of a feather.

a. Pinnately Lobed - Leaves are said to be pinnately lobed when the divisions are generally rounded and extending less than half-way to the midrib. (White Oak.) (Greek: lobes - lobe.)
b. Pinnately Fid - Leaves are said to be pinnately fid when the incisions are half-way to the midrib. (Maple, Red Oak.) (Latin: fidus - split.)
c. Pinnately Partite - Leaves are said to be pinnately partite when the incisions are more than half-way. (Shepherds Purse.) (Latin: partitus - divided.)
d. Pinnately Sect - Leaves are said to be pinnately sect when the incisions are almost to the midrib. (Field Poppy.) (Latin: secare - cut.)

17. When the Margins Have Deeper Incisions They Take on a Double Name - The Palmately - For the same reason Reticulate Multicostate leaves prefix the word palmati or palmately to the

term specifying each kind. Here the several portions of the leaf start from one point and diverge somewhat like the fingers, which start from the palm of the hand.

a. Palmately Lobed - Leaves are said to be palmately loved when the divisions are generally rounded and extending less than half-way to the midrib. (Ivy.)
b. Palmately Fid - Leaves are said to be palmately fid when the incisions are half-way to the midrib. (Black Currant.)
c. Palmately Partite - Leaves are said to be palmately partite when the incisions are more than half-way. (Hawthorn.)
d. Palmately Sect - Leaves are said to be palmately sect when the incisions reach almost to the midrib. (Monkshood.)

18. Dented Margins

a. Serrate - Dented margins of leaves are serrate when the divisions are pointed like the teeth of a saw and are directed towards the apex. (Deadnettle.) (Latin: serra - a saw.)
b. Crenate - Dented margins are said to be crenate when the incisions are rounded. (Ground Ivy.) (Latin: crena - notch.)
c. Dentate - Dented margins are said to be dentate when the divisions are pointed like teeth of a saw but are perpendicular to the margin, instead of being directed towards the apex - somewhat like the teeth in the mouth. (Guelder Rose.) (Latin: dens, dentis - tooth.)

19. Compound Leaves

a. Pinnate - Compound leaves are said to be Pinnate when the leaflets are disposed along the midrib. (Rose.) (Latin: pinna - feather.)
b. Palmate - Compound leaves are said to be palmate when the leaflets start at the same point. (Horse Chestnut.) (Latin: palma - palm of the hand.)

20. Pinnate Leaves

a. Paripinnate - Pinnate leaves are said to be paripinnate when they have two terminal leaflets. (Tamarind; Cassia.) (Latin: par, paris - equal.)
b. Imparipinnate - Pinnate leaves are said to be imparipinnate when they have one terminal leaflet. (Rose; Ash.) (Latin: in - not.)

21. Phyllotaxis - The arrangement of the leaves on the stem. (Greek: phyllon - leaf and taxis - arrangement).

a. Alternate - The arrangement is said to be alternate when one leaf is produced at each node and the leaves are not in a vertical line with each other. (Wallflower.) (Latin: alternate - to act by turns.)

b. Opposite - The arrangement is said to be opposite when two leaves spring from one node and face one another. (Privet.) (Latin: opponers - to oppose, from ‘ob’ - facing; positus - placed.)
c. Whorled - The arrangement is said to be whorled when more than two leaves are produced at a node. (Cross-leaved Heather.) (Middle English: wharle - the whorl of a spindle; - to go around.)

V. FLOWER

22. Parts of the Flower

a. The Flower - The flower consists of four groups of modified leaves.
b. The Calyx - The calyx is the outermost covering of the flower. It is usually a green envelope made up of sepals. (Latin: calyx - husk, shell, cup.)
c. The Corolla - The corolla is built up of petals. (Greek: petalos - outspread, flat) which are sometimes brightly coloured. (Latin: coronula - a little crown.)
d. The Androecium - The androecium consists of the aggregate of stames which form the male part of the flower and produces the pollen. (Greek: andros - male and oikos - house.)
e. The Gynoecium - Gynoecium or aggregate of Pistils (Greek: gyne - female and oikos - house) is the central part of the flower and consists of one or more transformed leaves which by folding together and uniting at the edges form closed receptacles called carpels. (Greek: karpos - fruit.) It is the female part of the flower and produces the ovules which mature into seeds.

23. Types of Calyx

a. Gamosepalous - The calyx is said to be gamosepalous when the sepals are partly or entirely united. (Deadnettle, Primrose.)
b. Polysepalous or Dialysepalous - The calyx is said to be polysepalous or dialysepalous when the sepas are distinct from one another. (Buttercup.) (Greek: polys - many.) (Latin: separare - to separate.) (Greek: dialein - separate.)

24. The Position of the Calyx - The position of the calyx in relation to the pistil varies. It may be:
a. Inferior or Hypogynous - The position of the calyx is said to be inferior or hypogynous when the calyx is fixed below the ovary. (Wallflower, Buttercup.)
b. Superior or Epigynous - The position of the calyx is said to be superior or epigynous when the calyx is fixed above the over. (Currant, Parsley.)

25. The Corolla

a. Gamopetalous - The corolla is called gamopetalous when the petals are united. (Primrose.)

b. Polypetalous - The corolla is called polypetalous when the petals are separate. (Buttercup, Wallflower.)

26. The Forms of Corolla - Gamopetalous

a. Spurred - The flower is said to be spurred when the corolla has developed a spur. (Larkspur, Nasturtium, Monkshood.) (Middle English: spure - a spur.)
b. Tubular - The flower is said to be tubular when the corolla is in the form of a tube. (Thistle.) (Latin: tubus - a tube.)
c. Imbutiform or Funnel Shaped - The flower is said to be imbutiform or funnel shaped when the corolla has the appearance of an inverted cone or of a funnel. (Convolvulus.) (Latin: imbutus - a funnel.)
d. Ligulate - The flower is said to beligulate when the corolla is tongue-like or strap shaped (as in the floret of the Dandelion.) (Latin: lingua - tongue.)

27. The Forms of a Corolla - Polypetalous

a. Cruciform - The flower is said to be cruciform when the corolla has four petals in the form of a cross, (Wallflower.) (Latin: Crux, crucis - cross.)
b. Rotate - The flower is said to be rotate when the petals are disposed like the spokes of a wheel. (Field Geranium, Stork's Bill)
c. Papilionaceous - The flower is said to be papilionaceous when the corolla has the form of a butterfly. (Pea, Gorse.) (Latin: papilio - butterfly.)

28. The Stamen - (Latin: stamen - thread.)
a. The Stamen
b. The Filament - The filament is the stalk to which the anther is attached. (Latin: filum - thread.).
c. The Anther - The anther is the part of the stamen which develops and contains the pollen. (Greek: antheros - flowery.)
d. The Pollen - The pollen is a fine dust, each grain consisting of a single cell. (Latin: pollen, pollinis - fine flour.)
29. The Pistil

a. The Pistil
b. The Ovary - The ovary is the enlarged part of the pistil at the base of the style. It contains the ovules which forms the seeds. (Latin: ovum - egg; ovarium - receptacle of eggs.)
c. The Style - The style is the pillar-like structure connecting the stigma and the ovary. (Greek: stylos - a pillar.)
d. The Stigma - The stiga is the area of the pistil (generally placed at its apex) which receives the pollen grains. Its surface is usually moist or viscid.

30. An Indefinite Inflorescence

a. A Spike - An indefinite inflorescence is called a spike when the pedicels are absent, so that the flowers are sessile along the axis. (Plantain, Bistort, Verbena, Willow, Wheat.) (Latin: spica - a wheat head.)
b. A Raceme - An idefinite inflorescence is called a raceme when the pedicels are present and the flowers are distributed along the elongated axis developing from below towards the apex. (Currant, Hyacinth.) (Latin: racemus - a bunch of grapes.)
c. A Panicle - An idenfinite inflorescence is called a panicle when the axis of the iflorescence branches and each branch forms a raceme. (Rhubarb, Oats.) (Latin: panicula - a tuft of plants).
d. A Corymb - An indefinite inflorescence is called a corymb when the pedicels are produced at different levels and are of different lengths ofrming a cluster which is convex or nearly level. (Candytuft, Cherry.)) (Ltin: corumbus - a cluster of flowers.)
e. A Simple Umbel - An indefinite inflorescence is called a simple umbel when the axis is greatly contracted so that the pedicels appear to spring from the same point and are all of equal length, forming a flat or rounded flower cluster. (Cowslip.) (Latin: umbella - a little shadow umbrella.)
f. A Compound Umbel - An indefinite inflorescence is called a compound umbel when each pedicel bears a smaller or secondary umbel. (Fool's Parsley, Carrot.)
g. A Capitulum - An indefinite inflorescence is called a capitulum when the axis is contracted and the flowers are sessile, forming a rounded or flat cluster. (Daisy, Dandelion, Clover.) (Latin: capitulum - a small head.)
h. A Spadix - A spadix is a spike with a fleshy or succulent axis bearing male and female flowers. It is unusually enclosed by a large leaf called spathe. (Arum or Cuckoo Pink.) (Latin: spadix - a palm branch, broken off with its fruit.)
i. A Catkin - The catkin is a crowded spike of small apetalous, unisexual flowers. Catkins are the male flowers of Oak, Hazel, Sweet Chestnut, Willow, Poplar and Birch. (Cat and kin from its resemblance to a cat's tail.)

VI. FRUIT

31. Classification of the Fruit Based on the Pericarp

a. Succulent Fruits - Fruits are called succulent when some portion or the whole of the pericarp is succulent. (Latin: succulentus - juicy, from sucus - juice.)
b. Dry Fruits - Fruits are called dry when the pericarp is hard and dry. (Middel English: druye - free from moisture.)

32. Succulent Fruits (Simple, Aggregate, Composite.)

a. The Drupe - The drupe is a simple fruit in which the epicarp and the mesocarp are succulent, but the endocarp extremely hard. (Cherry, Plum, Apricot, Walnut.) (Latin: drupa - overripe wrinkled olive.)
b. The Berry - The berry is a simple fruit in which the whole of the pericarp is soft and succulent. (Gooseberry, grape, currant, orange.) (Middle English - berie.)

c. The Pepo - The pepo is a hard-rinded berry belonging to the gourd family. It is a simple fruit. (Pumpkin, melon, cucumber, etc.)
d. The Pome - The pome is a simple fruit in which the fleshy part and the skin are formed from the thalamus. Imbedded in it are several cartilaginous, aper, or bony carpels. (Apple.) (Latin: Pomum - a fruit.)
e. Aggregate - A fruit is aggregate when the fruitlets are succulent, separated and each contains a seed. (Blackberry, Raspberry.)
f. Composite Fruits - Pineapple, Mulberry.

33. Dehiscent Fruits
a. A Capsule - A capsule is a dry fruit, usually many seeded and composed of two or more carpels which may open by pores, valves or teeth. (Poppy, Lily, Foxglove, Violet.) (Latin: capsula - a little box.)
b. A Siliqua - A siliqua is a fruitformd by two carpels. The two walls of the fruit separate and hang freely from the apex thus leaving free the structure made by the fusion of the two placentas to which the seeds are attached. (Wallflower, Rape, Mustard.) (Latin: siliqua - a pod.)
c. A Legume or Pod - A legume or pod is a gruit composed of a single carpel which dehisces along both the ventral and dorsal sides. (Pea, Vetch, Bean, Clover.) (Latin: legumen - a pod.)
d. A Follicle - A follicle is a fruit in which the single carpel or carpels open along one side only, usually the ventral side. (Columbine, Aconite, Larkspur.) (Latin: folliculus - a small bag.)

VII. SEEDS

34. Seeds - The seed of a flowering plant is the fertilized and ripened ovule. (Middle English: seed.) It consists of Spermoderm and Embryo.
a. The Spermoderm - The speroderm is the seed coat or outer covering which encloses an protects the embryo. (Greek: sperma - Seed; derma - skin.)
b. The Embryo - The embryo is the miniature plant.

35. Mono- and di- cotyledon - These are accompanied by either one or two masses of nourishing matter which are called seed leaves or Cotyledons. (Greek: Kotyledon - a cap-shaped hollow.)
a. Monocotyledon - Monocotyledon is the term applied when there is only one seed leaf present. (Maize, Coconut, Wheat.) (Greek: mono - one, alone.))
b. Dicotyledon - Dicotyledon is the term applied when there are two seed leaves present. (Bean.) (Greek: dis - two.)